

Mathematical Foundations of Image Synthesis

Kazi ASEF KABIR (2305084)

July 2026

1 Introduction

Image synthesis is the process of generating images from mathematical and algorithmic descriptions. Modern graphics systems rely on **precise mathematical models** and efficient computation. *Even small numerical errors* can result in visible artifacts in rendered images.

Typical formulations involve scalar values such as x , indexed parameters like I_{ij} , and exponential terms e^{x^2} . These elements are combined to simulate realistic lighting behavior.

2 Core Components

2.1 Rendering Pipelines

The rendering pipeline consists of the following stages:

- Scene Definition
- Camera Configuration
- Light Placement

Important Shading Model

- Diffuse Shading
- Specular Highlights

2.2 Execution Order

1. Input Processing
2. Ray Construction
 - (a) Primary Rays
 - (b) Secondary Rays
3. Intersection Evaluation
 - Object Hit test

3 Mathematical Modeling

Light interaction with surfaces can be expressed using the following equations.

$$L = I_{ij} \cdot \cos(\theta) + R^2 \tag{1}$$

$$k = \frac{ab}{cd}$$

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x + 1)^2 \end{aligned} \tag{2}$$

$$|y| = \begin{cases} y & \text{if } y \geq 0 \\ -y & \text{if } y < 0 \end{cases}$$

$$p(t) = \begin{cases} 0, & t < 0 \\ e^{-t}, & 0 \leq t < 3 \\ \ln(t - 2), & t \geq 3 \end{cases}$$

$$\mathcal{N}(x_i, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

4 Performance Comparison

Method	Accuracy	Time Complexity
Rasterization	Medium	O(n)
Ray Tracing	High	$O(n^2)$
	Very High	$O(n^3)$
Hybrid Method		$O(n^2)$

Table 1: Comparison of Image Synthesis Techniques

The results in Table 1 highlight the trade-off between accuracy and computational cost.

5 Visual Illustration

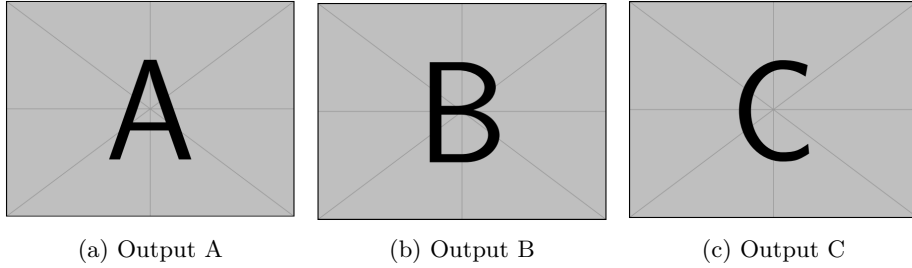


Figure 1: Comparison of three rendering outputs.

Figure 1 compares three different rendering outputs arranged in a single row.

6 Conclusion

Image synthesis combines mathematical precision, algorithmic design, and efficient implementation. As scene complexity increases, choosing the appropriate method becomes critical for balancing quality and performance.